

$$c_1 e^t + c_2 e^{-t}$$

$$y'' =$$

$$(c_1 e^t + c_2 e^{-t})$$

$$y' =$$

$$(c_1 e^t - c_2 e^{-t})$$

$\Rightarrow$

$$e^t (c_1 + m c_1 + n c_1) + e^{-t} (c_2 - m c_2 + n c_2) = 0.$$

$$1 + m + n = 0$$

$m + n = -1$

and

$$1 - m + n = 0$$